Repeated Games (Not in the Book)

ECON 5001
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Where We Left Off

Prisoners' Dilemma:

		Cooperate	Defect
Cooperate		2, 2	0, 3
Defect		3, 0	1, 1

Think of it as a price war (deck 03): Dos Hermanos vs. Pitabilities

- Cooperate = keep prices Normal. Defect = cut prices
- Same block, **every day**: a repeated game!

- Play once: D is strictly dominant
- Play repeatedly: Can we enforce Cooperation?

MobLab: “Prisoner’s Dilemma (Matrix)”

- MobLab time!
- Twice-repeated PD game
- Play the PD game twice with the same partner
- Repeat with a new partner
  - They can’t see how you played previously

Feel free to talk! And play it the way you’d actually play it.

Playing Twice Can Introduce New Nash Equilibria

Example

Player 1 \ Player 2	L	C	R
T	1, 1	5, 0	0, 0
M	0, 5	4, 4	0, 0
B	0, 0	0, 0	3, 3

Pure NE: _____

"Best" Outcome: _____

Example



Pure NE: (T, L) (B, R)

"Best" Outcome: (M, C) but not NE

Row → T

Col → L

Example

Player 1 \ Player 2	L	C	R
T	$\underline{1}, \underline{1}$	$\underline{5}, 0$	$0, 0$
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Pure NE: _____

"Best" Outcome: _____

Is (M, C) a NE? No

Example

Player 1 \ Player 2	L	C	R
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Pure NE: _____

"Best" Outcome: _____

Is (M, C) a NE? _____

Now play it **twice**:

- Period 1: play the game. Everyone sees what happened
- Period 2: play it again
- Total payoff $=$ period 1 payoff $+$ period 2 payoff

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Is (M, C) a NE? _____

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Can (M, C) happen in period 1???

A Strategy With a Threat

Player 1 \ Player 2	L	C	R
T	1, 1	5, 0	0, 0
M	0, 5	4, 4	0, 0
B	0, 0	0, 0	3, 3

Period 1: Play (M, C)

Period 2:

- If (M, C) happened: play (B, R)
- Otherwise: play (T, L)

Reward with the good NE. Punish with the bad NE

Check P1 in period 1:

- Conform (M): $4 + 3 = 7$
- Deviate (T): $5 + 1 = 6$

IS an SPNE

(T, L) is a credible punishment bc it's a NE of the 2nd game

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Check P1 in period 1:

- Conform (M): 4 + 3 = 7
- Deviate (T): 5 + 1 = 6

pays

(T,C) (T,C)

Gai: 0 + 1 = 1

Period 2? Playing a NE either way, no incentive to deviate

Every Period-2 Subgame

After period 1, there are 9 possible **histories**.

History: What's happened so far in the game $((T, L), (T, C), (M, B)$, etc.)

Period 2: Strategies can respond to the history ; need a plan for all 9 histories

After history (M, C): play (B, R)
After any other history: play (T, L)

SPNE: Solve via backward induction.

Every Period-2 Subgame

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Lesson: Period 2 can be used to **punish** & **reward** period 1 behavior

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After any other history: play (T, L)

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Lesson: Period 2 can be used to **punish** & **reward** period 1 behavior
BUT... what happens in Period 2 better be a NE. Otherwise, not credible

Formalizing Repeated Games

Stage Game vs. Supergame

- **Stage game** G: the game played each period
  - Our 3×3 game, or the PD
- **Supergame** G^T: play G in periods $t = 1, 2, \ldots, T$ ← T is # of periods/stages
  - Everyone sees all past play
  - T can be finite or ∞

Action: what you do in **one** period

- $a_i^t \in A_i$ (eg, M)
- Profile: $a^t = (a_1^t, a_2^t)$

actions

(T, L)

Strategy: a complete plan for G^T

- What to do in **every** period
- ...after **everything** that could happen

Histories

History at the start of period t: everything that happened before t

$$h_t = (a^1, a^2, \ldots, a^{t-1})$$

- $h_1 = \emptyset$ (nothing has happened yet)
- $H_t =$ set of all possible histories at the start of period t

Ex: Suppose the 3×3 game is played 4 times:

- $h_2 = (M, C)$ is one history at start of period 2
- $h_3 = ((M, C), (T, L))$ is a possible history at start of period 3
- $h_4 = ((M, C), (T, L), (M, C))$ is a possible period-4 history
- $|H_2| =$ 9 $\quad |H_3| =$ 9 × 9 $\quad |H_4| =$ 9 × 9 × 9

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- $|H_2| =$ _____ $|H_3| =$ _____ $|H_4| =$ _____

Each history starts a **subgame** (it's a singleton node in the tree)

Pure Strategy in a Repeated Game

Definition
A **pure strategy** for Player i is a function s_i that maps every history to an action:
$s_i(h_t) \in A_i$ for every h_t in every H_t

history

Back to our 2-period 3×3 game. Our threat strategy for P1:

$s_1(\emptyset) =$ M

$s_1((M,C)) =$ B when $h_2 = (M,C)$ ← in period 1

$s_1(h_2) =$ T for every other $h_2 \neq (M,C)$

$s_2(\emptyset) = C$

$s_2((m_1,c)) = R$

$s_2(h_2) = L$ any $h_2 \neq (m_1,c)$

3-period version

$s_1((m_1,c),(m_1,c)) = B$

$s_1(h_3) = T$ for any $h_3 \neq ((m_1,c),(m_1,c))$

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\begin{aligned}
s_1(\emptyset) &= _____ \\
s_1\big((M, C)\big) &= _____ \quad \text{when } h_2 = (M, C) \\
s_1(h_2) &= _____ \quad \text{for every other } h_2 \neq (M, C)
\end{aligned}
$$

P1 moves after how many different histories? $1 + \underline{9} = \underline{10}$
of pure strategies:

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Imagine chess...

Two Kinds of Equilibrium

Stage-game NE: a NE of G (one period)

- (T, L) and (B, R) in our example

Equilibrium of the supergame: a NE or SPNE of G^T

- A profile of **strategies** (s_1, s_2), not actions
- SPNE: a NE in the subgame after **every** history

– Last period: SPNE plays NE of stage game

Ex: our threat strategies are a SPNE of G^2

- Period-1 play of (M, C) is part of a SPNE
- Even though (M, C) is **not** a stage-game NE!

"(M, C) then (B, R)" is not a complete answer

Stage NE Every Period Is Always Safe

Theorem
Let a^* be a NE of the stage game G. Then "play a^* in every period, no matter what happened" is a SPNE of G^T.

Same for any **sequence** of stage NE that ignores history: $a^{1*}, a^{2*}, \ldots$

Why?

- Future play doesn't depend on what I do today
- Only today's payoff matters
- a^* is a stage NE $\Rightarrow$ no profitable deviation today

Thus: $SPNE(G^T)$ is never empty if G has a NE

Twice-Repeated Prisoners' Dilemma

Two-Period PD

	Cooperate	Defect
Cooperate	2, 2	0, 3
Defect	3, 0	1, 1

Price war over two days

Try a punishment strategy:

- Period 1: (C, C)
- Period 2: reward if both Cooperated, punish otherwise

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Reward with what?
Punish with what?

Two-Period PD

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Punish with what?

Period 2: every subgame is a one-shot PD $\Rightarrow$ _____

Period 1: period 2 is the same no matter what $\Rightarrow$ _____

Two-Period PD

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Period 1: period 2 is the same no matter what $\Rightarrow$ _____

Problem: only **one** stage-game NE: (D, D). No credible punishment/reward.

Any Finite Number of Periods

- Period T: last period. Play _____
- Period $T-1$: can't affect period T play. Play _____
- Period $T-2$: ...
- ...
- Period 1: _____

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Theorem
If G has a unique NE and T is finite, the only SPNE of G^T is to play that NE every period.

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- Period $T-2$: ...
- ...
- Period 1: _____

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If G has a unique NE and T is finite, the only SPNE of G^T is to play that NE every period.

The game **unravels** from the end.

But In the Lab...

Lab result: people **do** cooperate in finitely repeated PDs

- Lots of cooperation early
- Collapses in the last few rounds (“end-game effect”)
- With experience, the collapse starts earlier

More cooperation when players aren’t sure the other is purely selfish

- A little doubt about the other’s **type**...
- We’ll see this in Chapter 15!

Source: Caplan, *A Practicum in Behavioral Economics*, §3, p. 176.
Selten & Stoecker (1986, *JEBO*); Andreoni & Miller (1993, *EJ*).

Infinitely Repeated Games

No Last Period

What if the game doesn't have a known end date?

1. $T = \infty$, or
2. The game ends at a random time (more realistic)

Problem 1: Can't apply backward induction/SPNE

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Problem 1: Can't apply backward induction/SPNE

Problem 2: payoffs add up to infinity

- Always Cooperate: $2 + 2 + 2 + \cdots = \infty$
- Defect once, then get Defected on forever: $3 + 1 + 1 + \cdots = \infty$

Which would you prefer?
Can't compare ∞ to ∞!

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Which would you prefer?
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Fix: future payoffs count **less** than today's

Discounting: Money Now vs. Later

Interest rate $r = 10\%$:

- At 10% interest, \$90.91 today grows to $\$90.91 \times 1.10 = \$100$ next year
- So \$100 next year is worth _____ today
- \$82.64 today grows to $\$82.64 \times 1.10 = \$90.91$ next year, then to \$100 in 2 years
- So \$100 in two years is worth _____ today

General: What amount today will grow to \$100 in t periods?

$$
\begin{aligned}
x\,(1+r)^t &= 100 \\
x &= \left(\frac{1}{1+r}\right)^t 100 \\
x &= \delta^t 100
\end{aligned}
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Discount factor δ: how much we discount future payoffs per period: $\delta = 1/(1+r)$

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Patient: don't discount much. δ near 1

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Discount factor δ: how much we discount future payoffs per period: $\delta = 1/(1+r)$
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Patient: don't discount much. δ near 1
Impatient: discount a lot. δ near 0

Utility of a Stream of Payoffs

"Stream" of payoffs: u_i^1 today, u_i^2 next period, u_i^3 after that, *etc.*

"Exponential" discounting: add them up, discounting each period by another δ.
Total value of the stream of payoffs:

$$U_i = u_i^1 + \delta u_i^2 + \delta^2 u_i^3 + \cdots = \sum_{t=1}^{\infty} \delta^{t-1} u_i^t$$

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Fact: if $0 \leq \delta < 1$, this is **finite**

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Important Example: Suppose the stream pays x every period forever:

$$U_i = x + \delta x + \delta^2 x + \delta^3 x + \cdots \qquad \delta U_i = _____$$

$$U_i - \delta U_i = (1-\delta)U_i = ____ \quad \Rightarrow \quad U_i = _____$$

The Infinitely Repeated PD

	Cooperate	Defect
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One SPNE we already know:

- Always Defect, no matter what
- Why? _____

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Powerful strategy: **Grim Trigger:**

- Period 1: play Cooperate
- Keep Cooperating as long as **everyone** always has
- Once anyone Defects: you Defect **forever**
- Price war: Normal prices until someone cuts. Then cut forever

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- Price war: Normal prices until someone cuts. Then cut forever

As a function of histories: $s_i(h_t) = C$ if h_t is all (C, C), otherwise $s_i(h_t) = D$

Is Grim Trigger a SPNE?

Two kinds of subgames:

1. Someone already Defected

- Both Defect forever: stage NE every period
- Theorem $\Rightarrow$ no profitable deviation ✓

2. Everyone always Cooperated

- Conform: $2 + 2\delta + 2\delta^2 + \cdots =$ _____
- Deviate (Defect today, triggering the Grim Trigger response tomorrow):
  - Today: _____
  - Then punished forever starting tomorrow: $\delta 1 + \delta^2 1 + \delta^3 1 + \cdots$
  - Total payoff from this plan:

    $3 + \delta 1 + \delta^2 1 + \delta^3 1 + \cdots$

    $2 + \left(1 + \delta 1 + \delta^2 1 + \delta^3 1 + \cdots\right)$

    $2 + \frac{1}{1-\delta} 1$

The Threshold

In words: cooperating forever must beat grabbing 3 today and getting punished forever

Price war: is one day of extra sales from a price cut worth a price war forever?

$$\underbrace{2\frac{1}{1-\delta}}_{\text{cooperate forever}} \geq \underbrace{2+\frac{1}{1-\delta}1}_{\text{grab today, punished forever}}$$

Solve for δ:

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Solve for δ:

$$\begin{aligned} 2 &\geq 2(1-\delta)+1 \\ 1 &\geq 2(1-\delta) \\ 1 &\geq 2-2\delta \\ 2\delta &\geq 1 \\ \delta &\geq 1/2 \end{aligned}$$

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Result: Grim Trigger is a SPNE $\iff \delta \geq$ _____. In other words, if they're patient enough

Other Strategies

- **Tit-for-Tat:** start with Cooperate, then copy what they did last period
  - Won Axelrod's computer tournaments (1980)
  - Nice, retaliatory, forgiving
- **Always Cooperate**, no matter what
  - Is playing "Cooperate at every history" a SPNE? _____
- **Always Defect**: always a SPNE

Not in the book. Axelrod, *The Evolution of Cooperation* (1984). Tit-for-Tat was submitted by Anatol Rapoport.

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Lesson: cooperation needs a credible **threat**

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This Is Everywhere

Repeated PDs, with cooperate / defect in each:

- **Cartels:** OPEC members: stick to quota / pump more
- **Price wars:** airlines on a route: hold fares / slash fares
- **Trade:** countries: keep tariffs low / raise tariffs
- **Roommates:** do your chores / let them pile up
- **Online sellers:** ship what you promised / ship junk
- **WWI trenches:** live and let live / non-stop combat

In each: the **future** keeps people honest today

Trenches: “live and let live,” Axelrod (1984). Many historians disagree.

But Those Are More Complicated

Real versions add:

- **Imperfect monitoring:** you can't directly see what your opponent did
  - e.g., you don't see their price cut, but you see a drop in your demand
- **Many players:** whom do you punish?
- **Entry:** new players who weren't part of the original agreement
- **Renegotiation:** after a price war, both want to "let bygones be bygones"

Core tension survives: today's gain vs. tomorrow's punishment.
And, is tomorrow's punishment even credible?

How Much Can Repetition Do?

Minmax Payoff

In words: the worst the others can hold you to, if you always best respond

$$v_i = \underbrace{\min_{a_{-i}}}_{\text{others hurt you}} \quad \underbrace{\max_{a_i} u_i(a_i, a_{-i})}_{\text{you best respond}}$$

Ex: the PD

- If P2 plays Cooperate, P1's best: _____
- If P2 plays Defect, P1's best: _____
- So $v_1 =$ _____ (same for P2)

Price war: a rival who cuts prices forever holds you to 1 per day

Others could also mix to hold you down. We'll ignore that.

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Ex: the PD

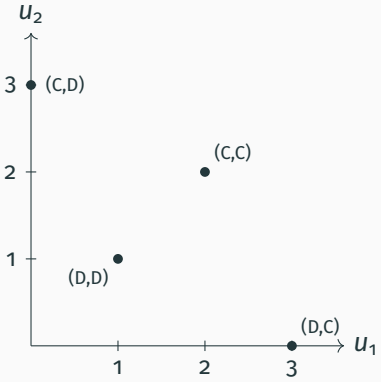
- If P2 plays Cooperate, P1's best: _____
- If P2 plays Defect, P1's best: _____
- So $v_1 =$ _____ (same for P2)

Price war: a rival who cuts prices forever holds you to 1 per day

Fact: You can always get at least v_i. No equilibrium can give you less

Others could also mix to hold you down. We'll ignore that.

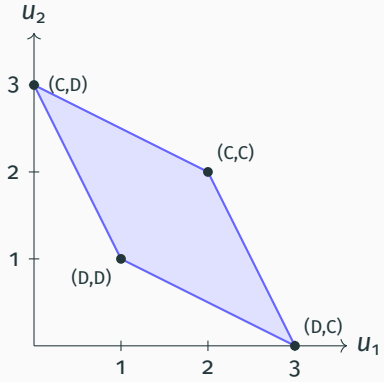
Feasible Payoffs



Per-period average payoffs

Average payoff: $(1-\delta)U_i$. Getting x forever gives an average of x.

Feasible Payoffs

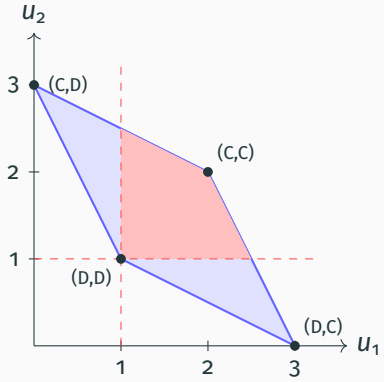


Per-period average payoffs

Feasible: any mix of the four outcomes (take turns, etc.)

Average payoff: $(1-\delta)U_i$. Getting x forever gives an average of x.

Feasible Payoffs



Per-period average payoffs

Feasible: any mix of the four outcomes (take turns, etc.)

Above minmax: both get more than $v_i = 1$

Average payoff: $(1 - \delta)U_i$. Getting x forever gives an average of x.

The Folk Theorem

Folk Theorem (stated without proof)
In an infinitely repeated game, **any** feasible payoff giving every player more than their minmax v_i is the average payoff of some SPNE, if δ is close enough to 1.

- "Folk": everyone knew it before anyone published it
- PD: anything in the red region. Including (2,2): (C, C) forever!
- Also including _____ (barely better than always Defect)

Not in the book. Friedman (1971); Fudenberg & Maskin (1986, *Econometrica*). Some technical conditions apply.

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- **When** cooperation is possible: patience, no known end, seeing what others did
- **Comparative statics:** higher $\delta \Rightarrow$ more cooperation. The lab agrees
- Explains **institutions**: reviews, credit scores, cartel meetings

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New question: which equilibrium? Focal points, talking, culture...

Let’s Work One Out

Gas on Lane Avenue

Shell and Speedway face each other at the same intersection. Every day each posts a **High** or **Low** price. Shell has the bigger brand

		Speedway: High	Speedway: Low
Shell	High	6, 4	1, 6
Shell	Low	8, 1	3, 2

Shell picks the row, Speedway the column. Payoffs (Shell, Speedway) in $100s per day

They play forever, discount factor δ

Gas on Lane Avenue

1. Find the stage-game NE and each station's minmax payoff

2. For which δ is Grim Trigger (High until anyone goes Low) a SPNE? Whose incentive binds?

3. Now a price cut is only noticed **one day late**. New threshold?

4. Folk theorem: can average payoffs $(4, 3)$ be a SPNE for δ near 1? What about $(2, 5)$?

What Have We Learned, Charlie Brown?

- **Stage game** vs. **supergame**. Actions vs. strategies
- Strategy: a map from **histories** to actions
- Stage NE every period: always a SPNE
- Multiple stage NE: reward and punish, even with T finite
- Unique stage NE and T finite: **unraveling**
- $T = \infty$: discount with δ. Grim Trigger works if δ is big enough
- **Folk theorem**: anything above minmax, if patient enough

Reading & Practice

- Not in Bonanno. These slides are the reading
- Optional: Osborne, *An Introduction to Game Theory*, Chapters 14–15
- Optional: Caplan, §3, p. 176 (finitely repeated PD)
- Practice: redo the PD with $(C, C) = (2.5, 2.5)$. New threshold?